EARTHPRESS · PRODUCT REQUIREMENTS AND SPECIFICATIONS

EarthPress product requirements and specifications

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	Kit 1.8 migration of EPR-REQ-001 v0.8 (R1 to R14 become PR-001 to PR-014, old number in "Was"); PR-015 and PR-016 added; traced to EPR-URS-001; status from EPR-CAL-001 v0.7 (PR-007 not met at 203.8 kg; PR-013 USD 542; PR-016 met after fix F1)

EarthPress product requirements and specifications

What the press must do, each requirement traced to a user need in EPR-URS-001 (links in EPR-RTM-001). This document replaces EPR-REQ-001 (v0.8 of 2026-10-02): every requirement keeps its target, verification and status, with its old number in "Was". Status is from the calculation note EPR-CAL-001 v0.7 for the current design (option A, the lever and toggle press). On paper 8 of the 16 requirements are met, two are at risk, one is **not met** (PR-007, mass), one is over its value-engineering target (PR-013, cost) and four can only be verified at TRL 4. Targets remain proposals for review with a co-design partner (EPR-PRB-001). Method: STANDARDS section 22.

The **reference block** is 290 x 140 x 90 mm, about 3.65 L and 6.94 kg dry, from a sandy clay loam stabilized with 5 % cement by dry mass (decided by Amish, 2026-09-25; EPR-DDR-001 items 3 to 5, EPR-DDR-002).

Table 1. Product requirements and specifications. Status from EPR-CAL-001 v0.7 (2026-10-25).

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001	Performance	Block dimensions	Blocks shall be 290 x 140 mm on plan and 90 mm high, checked with the block gauge	290 x 140; 90	±2 on plan; ±3 on height	mm	Analysis (CAL-001 sections 5 and 6), TRL 3; gauge checks on a production run, TRL 4	Met on paper: 90.0 mm at the crank stop with the fill within 0 to +2.7 %; mold bulge 0.08 mm	R1
PR-002	UR-002	Performance	Compaction pressure	The press shall reach 2.0 MPa or more on the block face at the end of the stroke	2.0	min	MPa	Analysis (CAL-001 sections 3 to 5), TRL 3; load cell, TRL 4	**At risk**: 2.00 MPa with two operators and a weighed fill; one operator at 500 N stalls at 0.53 MPa; rests on the assumed soil law (STM S6)	R2
PR-003	UR-002, UR-016	Performance	Block strength with suitable soil	Blocks from suitable soil shall reach 4 MPa dry and 2 MPa wet at 28 days (Auroville class B), and in all cases 2.1 MPa (300 psi) as in 14.7.4 NMAC	4; 2; 2.1	min	MPa	Literature, TRL 3; block tests, TRL 4	To verify at TRL 4	R3
PR-004	UR-003	Performance	Output	A crew of four shall make 300 or more blocks in an 8 h day, including mixing	300	min	blocks a day	Analysis (CAL-001 section 9), TRL 3; timed trials, TRL 4	**At risk**: 296 a day on an assumed 85 s cycle (shortfall accepted by Amish, 2026-09-25)	R4
PR-005	UR-004	Usability	Operator force and grip height	The peak pull per operator at the grip shall be 500 N or less, with the grip between	500; 0.8 to 1.9; 1.0	max; band; min	N; m; m	Analysis (CAL-001 sections 3, 4 and 4b), TRL 3;	Met on paper with two operators: 334 N each (base soil), 405 N each	R5

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				0.8 and 1.9 m above the ground and the peak at 1.0 m or higher				force at the grip, TRL 4	(softer design soil); grip 0.86 to 1.89 m; peak at 1.16 m; eject 469 N for one	
PR-006	UR-005	Physical architecture	Garage-buildable	The press shall be built with a stick welder, angle grinder, drill press and hand tools; turned parts only as bought pins, shafts and bushes	0	max	machined parts made in the shop	Inspection of the model, drawings and build plan (BLD-001), TRL 3	Met by design	R6
PR-007	UR-006	Packaging and transport	Movable	The complete press, including the guard, bolts and bushes, shall weigh 200 kg or less; no piece over 50 kg after removing pins and bolts; it shall fit a 1.5 m pickup bed	200; 50; 1.5	max	kg; kg; m	Analysis (CAL-001 section 8, kit 1.9.1 model), TRL 3; weighing, TRL 4	**Not met**: 203.8 kg with every bush, nut, washer and weld bead (3.8 kg over; decision D2, proposed, awaiting Amish); heaviest piece 44.5 kg (met); plan 1.28 x 0.58 m (met)	R7
PR-008	UR-007, UR-015	Materials	Durability	The press shall make 50,000 blocks before mold liners or bushes need replacing; all pins in replaceable bushes; a grease point on every pivot	50,000	min	blocks	Analysis (CAL-001 section 6, fatigue), TRL 3; endurance log, TRL 4	To verify at TRL 4: welded range 89 MPa against 243 MPa, pin range 234 MPa against about 350 MPa; wear needs soil data	R8
PR-009	UR-008	Usability	Soil test kit		1; 24	max	h			R9

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				Field tests (jar sedimentation, shrinkage box, ribbon and drop tests) shall give a go or no-go and a starting cement content, a first screen within 1 h and a full result within 24 h				Inspection of the written procedure (BLD-001), TRL 3; comparison with laboratory grading, TRL 4	Met by design; accuracy to verify at TRL 4	
PR-010	UR-010	Materials	Low stabilizer use	Blocks shall reach PR-003 with 8 % cement by dry mass or less; 5 % is the design value	8	max	% of dry mass	Literature, TRL 3; block tests, TRL 4	To verify at TRL 4 (design at 5 %, 0.33 kg a block)	R10
PR-011	UR-011	Materials	Embodied carbon	Cradle-to-gate CO2 of the walling material shall be 25 % or less of a fired-brick wall of equal volume	25	max	%	Analysis (CAL-001 section 10, cited factors), TRL 3	Met on paper: 7.6 % (0.35 t against 4.55 t for a small house)	R11
PR-012	UR-012	Safety	Safe operation	The lever shall not fall freely when released under load; the lid latch shall hold at least 1.5 times the peak block force; pinch points between lever and frame shall be guarded or have 100 mm or more clearance	1.5; 100	min	x 81.2 kN; mm	Analysis and design review (CAL-001 section 6), TRL 3; proof-load test, TRL 4	Met on paper: rest catch and end pawl; lid and latch rated 173 kN; linkage guard	R12
PR-013	UR-009	Cost	Affordable		500	max	USD			R13

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				Press and soil kit parts shall cost at or below the value-engineering target, excluding labor and cement				Analysis of the priced BOM (`bom/bom.csv`), TRL 3	**Over target**: USD 542 (indicative), USD 42 over; savings in EPR-DEC-001	
PR-014	UR-013	Usability	Open and documented	CERN-OHL-S drawings, BOM and build plan shall let a welder build the press without contacting the author	1	min	outside build review	Review by an outside fabricator, TRL 4	To verify at TRL 4	R14
PR-015	UR-014	Usability	Human powered	The press shall need no electricity, fuel or hydraulic fluid to press or eject a block	0	max	powered parts	Inspection of the model and BOM, TRL 3	Met by design	
PR-016	UR-012, UR-007	Safety	Structure at the design load	Every member in the load path shall stay below yield at the 173 kN design load (1,000 N at the grip at the crank stop)	1.0	min	safety factor on yield	Analysis (CAL-001 sections 6 and 6b), TRL 3; proof-load test, TRL 4	Met on paper after fix F1 (EPR-DEC-001): lowest factor 1.29; the 35 mm C45 pins did not meet it	

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Requirements not met or at risk

- **PR-007 not met:** 203.8 kg against 200 kg. The earlier 199.996 kg left out the bushes, nuts, washers and weld metal, which the kit 1.9.1 model counts, and fix F1 adds about 1.2 kg. Options in EPR-DEC-001 D2 (proposed, awaiting Amish).
- **PR-013 over target:** USD 542 against the USD 500 value-engineering target (a hypothetical control target, `budget_usd` unchanged).

- **PR-002 at risk:** met only with two operators, a weighed fill and the assumed soil stiffness.
- **PR-004 at risk:** 296 against 300 blocks a day on an assumed cycle; the shortfall was accepted by Amish on 2026-09-25 (EPR-DDR-002).
- PR-003, PR-008, PR-010 and PR-014 can only be verified at TRL 4, which is on hold.

Revisions accepted (EPR-DDR-002)

Decided by Amish, 2026-09-25: go with recommendation. R5 (now PR-005) was a peak pull of 500 N or less with the grip between 0.9 and 1.6 m; R7 (PR-007) was 140 kg, then 190 kg, and has been 200 kg including the guard and bolts since 2026-10-02 (EPR-DDR-003 A1); R13 (PR-013) was USD 450 and has been USD 500 since 2026-09-25.

Assumptions

- Compaction pressure of 2 to 4 MPa and about 5 % cement follow Auroville Earth Institute practice (Auroville Earth Institute).
- Dry block density about 1,900 kg/m³ (estimate), giving about 6.9 kg per dry block.
- One operator can pull 500 N at the grip for short strokes, using body weight; two can pull 1,000 N on the T-handle. To be checked with users.
- Soil pressure rises tenfold over the last 14 mm of the stroke (range 10 to 20 mm); to be measured with partner soils (EPR-STM-001 S6).
- A crew of four covers soil sieving, mixing, pressing and stacking.